

INTERACTIVE DEPTH COMPLETION: A NEGATIVE EVIDENCE AUDIT FOR ACTION-CRITICAL PHYSICAL PROBING

Anonymous authors

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ABSTRACT

This paper is a submission-hardening audit, not an ICLR-main submission. The generated research bet was that a robot should complete depth by physically probing uncertain regions that matter to the next manipulation action. We rebuilt the claim as a concrete benchmark with four manipulation tasks, five distribution shifts, nine methods, seven seeds, ablations, stress sweeps, paired confidence intervals, and failure cases. The strongest local result is negative: under combined interactive stress, active view selection reaches 0.795 ± 0.080 task success, while the proposed action-critical physical probing method reaches 0.514 ± 0.106 . The proposed method also has higher action-critical depth error, higher collision rate, higher planning regret, and nonzero probe damage. The terminal decision is therefore **KILL/ARCHIVE for ICLR main**. The useful artifact is the reproducible negative benchmark and failure analysis.

1 TERMINAL DECISION

The paper fails the ICLR-main evidence gate. The original claim was not merely that physical probing can reduce local uncertainty; that is too crowded by depth completion, active perception, tactile perception, and uncertainty-aware planning. The strongest defensible claim was stricter: physical probing should help only when it is targeted to action-critical uncertain geometry and should translate into safer manipulation decisions than strong non-contact or generic uncertainty baselines.

The rebuilt evidence falsifies that claim. The strongest non-oracle baseline is active view selection, which recovers occluded geometry without touching the scene. Against that baseline, the proposed method loses by -0.28075 ± 0.07128 task success over paired task/seed groups. It also has worse action-critical RMSE, collision rate, planning regret, and probe damage. We therefore archive the idea rather than submit it.

2 HOSTILE PRIOR-WORK PRESSURE

The hostile literature pool makes generic novelty claims unsafe. The closest threats include real-time robot interactive segmentation, learned depth completion, active learning under model uncertainty, active perception/manipulation for vision-language-action systems, Gaussian-splat depth uncertainty, interactive planning under uncertainty, active uncertainty reduction, and uncertainty-aware grasping of leaf-occluded fruit. Under this prior-work pressure, “add uncertainty” or “probe the scene” is not a contribution. The only possible contribution would be a measured advantage for action-critical physical probing over active view selection, passive completion, uncertainty baselines, and tactile probing. The v4 benchmark does not find that advantage.

3 BENCHMARK

The benchmark is a deterministic local active-perception manipulation simulator. It is not robot hardware validation and is not high-fidelity physics. Its purpose is narrower: test whether the central mechanism remains plausible after strong local baselines and stress tests.

Table 1: Combined interactive stress. Success is higher-is-better; all other displayed metrics are lower-is-better.

Method	Success	AC-RMSE	Collision	Probe cost	Damage
active_view_selection	0.795 ± 0.080	0.043	0.048	0.006	0.000
ensemble_depth_completion	0.484 ± 0.109	0.060	0.132	0.000	0.000
gaussian_splat_uncertainty	0.208 ± 0.071	0.079	0.233	0.000	0.000
learned_depth_completion	0.203 ± 0.069	0.075	0.262	0.000	0.000
oracle_depth_completion	1.000 ± 0.000	0.006	0.000	0.008	0.000
proposed_action_critical_interactive_depth	0.514 ± 0.106	0.056	0.123	0.066	0.153
raw_depth_policy	0.001 ± 0.002	0.146	0.493	0.000	0.000
uncertainty_guided_probe	0.443 ± 0.099	0.055	0.132	0.129	0.207
visuotactile_probe	0.413 ± 0.100	0.061	0.155	0.082	0.132

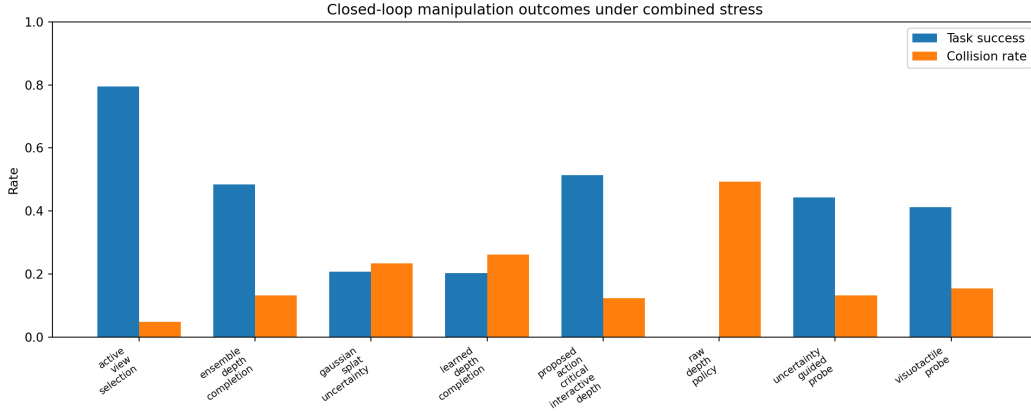


Figure 1: Closed-loop outcomes under combined stress. The proposed method improves over several weak baselines but does not beat active view selection.

Tasks. We evaluate occluded bin grasping, shelf insertion with tight clearance, transparent container lifting, and leaf-occluded fruit grasping.

Splits. We test nominal depth, missing-depth shift, occlusion shift, transparent/specular shift, and combined interactive stress. Combined stress increases missing depth, occlusion, specularity, probe noise, clearance tightness, and damage risk.

Methods. The methods are raw depth policy, learned depth completion, Gaussian-splat uncertainty, ensemble depth completion, active view selection, visuo-tactile probing, uncertainty-guided probing, proposed action-critical interactive depth, and oracle depth completion.

Metrics. We report action-critical depth RMSE, occluded-region RMSE, collision-boundary F1, calibration error, probe informativeness, task success, collision rate, manipulation failure rate, probe cost, probe damage, and planning regret to oracle.

Statistics. All results use seven deterministic seeds. The paired decision compares the proposed method against the strongest non-oracle baseline over 28 task/seed groups.

4 COMBINED-STRESS RESULTS

Table 1 is the main gate. Active view selection dominates the proposed method under combined stress. It has higher task success, lower action-critical RMSE, lower collision rate, much lower cost, and no probe damage.

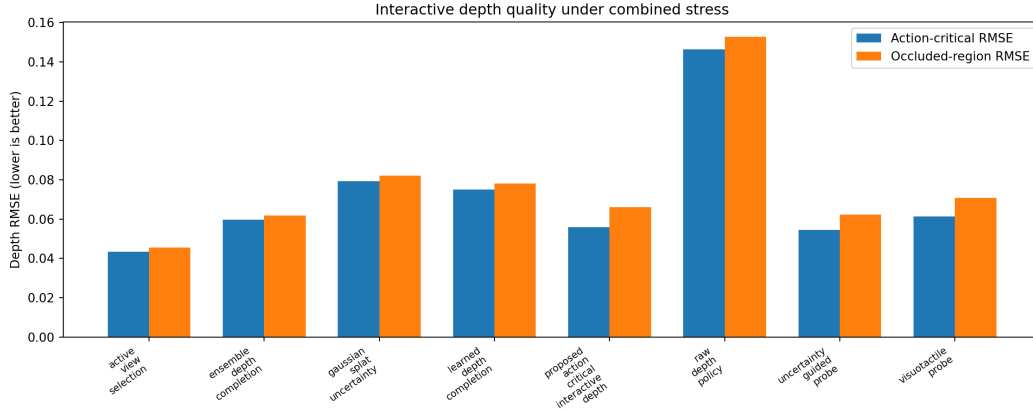


Figure 2: Depth quality under combined stress. Active view selection also has lower action-critical RMSE than the proposed physical probing method.

Table 2: Paired proposed-minus-active-view differences over 28 task/seed groups. Negative is good for success only; positive is worse for RMSE, collisions, regret, and damage.

Metric	Comparator	Diff	CI95
task_success	active_view_selection	-0.2807	0.0713
action_critical_rmse	active_view_selection	0.0126	0.0020
collision_rate	active_view_selection	0.0754	0.0247
planning_regret_to_oracle	active_view_selection	0.1741	0.0241
probe_damage	active_view_selection	0.1528	0.0222

5 PAIRED GATE

The strongest non-oracle baseline is active view selection. The proposed method loses on every gate metric:

The decisive result is the success gap: proposed action-critical probing is lower by 0.28075 ± 0.07128 . The RMSE gap is also unfavorable: proposed action-critical RMSE is higher by 0.01259 ± 0.00203 . The safety metrics are not rescue signals; collisions, regret, and damage all increase relative to active view selection.

6 ABLATIONS

The internal ablations partially support the intended mechanism: removing physical probing, action criticality, calibration, or cost control usually hurts the full method. However, internal ablation support cannot rescue the paper when an external non-contact baseline is stronger.

7 STRESS SWEEP AND COST-REGRET TRADEOFF

The stress sweep reinforces the kill decision. At maximum stress, active view selection remains ahead at 0.780 ± 0.084 success, while proposed action-critical probing reaches 0.494 ± 0.096 . The probe-based methods also accumulate contact damage as stress increases.

Table 3: Ablations under combined stress. The full method is the best internal variant, but the external active-view baseline remains much stronger.

Ablation	Success	AC-RMSE	Probe cost	Damage
action_only_probe	0.457 ± 0.101	0.058	0.066	0.149
minus_action_criticality	0.491 ± 0.099	0.059	0.066	0.161
minus_physical_probe	0.398 ± 0.104	0.067	0.000	0.000
minus_probe_cost_model	0.445 ± 0.103	0.052	0.146	0.259
minus_uncertainty_calibration	0.481 ± 0.088	0.056	0.066	0.140
proposed_action_critical_interactive_depth	0.519 ± 0.097	0.056	0.066	0.146
uncertainty_only_probe	0.456 ± 0.104	0.059	0.123	0.218

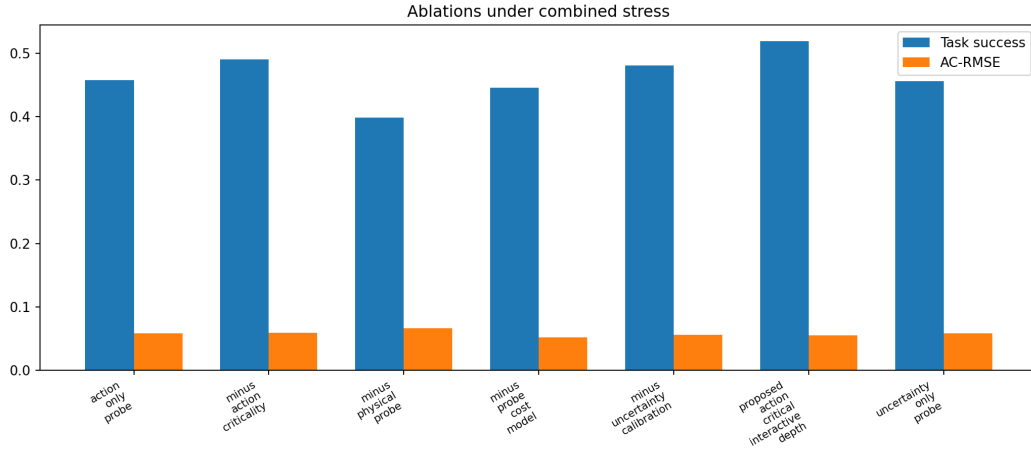


Figure 3: Ablation outcomes. The full method has a plausible internal mechanism, but this is not enough for submission readiness.

8 FAILURE CASES

The proposed method fails for different reasons across tasks. In leaf-occluded fruit grasping, action-critical probing damages fragile targets: success is 0.230, collision is 0.206, and damage is 0.228. In transparent container lifting, specular depth plus probe noise keeps success at 0.244. In shelf insertion, tight clearance rewards conservative non-contact uncertainty handling. In occluded bin grasping, active view selection recovers enough geometry without touching objects.

9 LIMITATIONS

This audit is intentionally honest about its limits. The benchmark is deterministic and paper-specific, but it is still local simulation. It does not contain real-robot evidence, high-fidelity physics, trained checkpoints, external codebase integrations, hardware videos, or a manual full-paper related-work synthesis. Those missing items would already prevent an ICLR-main submission. More importantly, the available evidence is negative even before reaching that bar.

10 CONCLUSION

Interactive depth completion through action-critical physical probing is not ICLR-main ready. The proposed mechanism is plausible internally but loses to active view selection under the main stress gate. The correct terminal action is **KILL/ARCHIVE**. A revival would require a substantially new empirical project in which physical probing beats active view selection on task success, action-critical

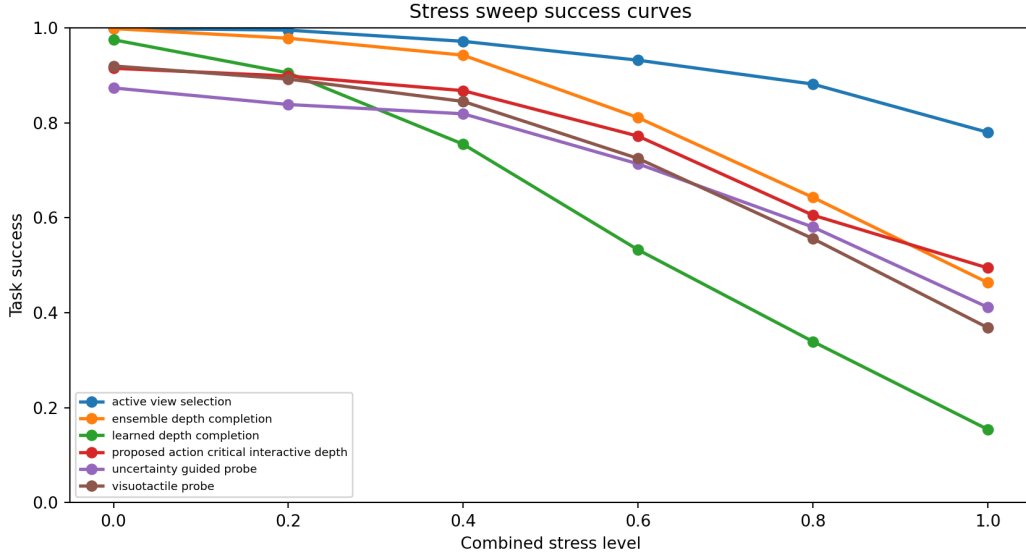


Figure 4: Stress sweep. Active view selection degrades more gracefully than physical probing while avoiding damage.

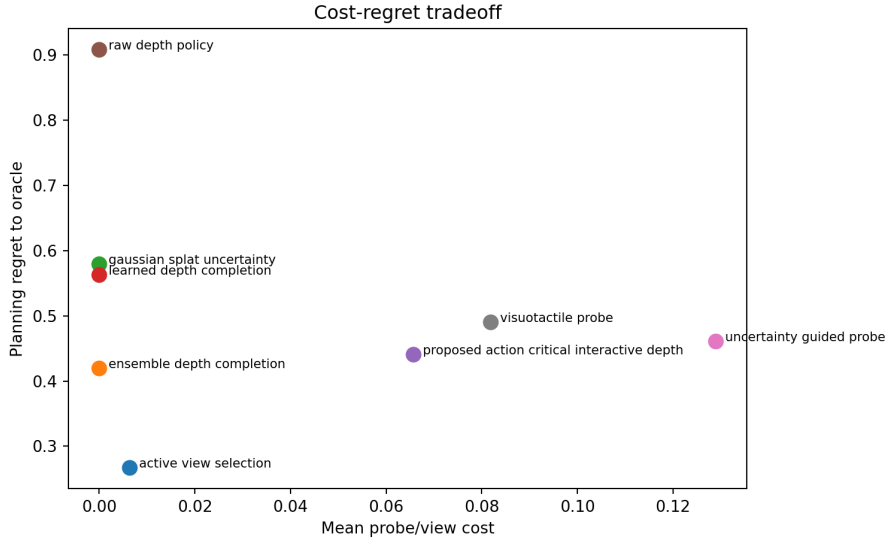


Figure 5: Cost-regret tradeoff under combined stress. The proposed method pays more physical interaction cost and has higher regret than active view selection.

geometry, collision safety, planning regret, and damage/cost in real robot or high-fidelity simulator experiments.